AnnaSaathi — Product Requirements Document

Team SparkForge · Triwizathon 2026 · Problem Statement 5 · Oct 24, 2026

1. Overview

AnnaSaathi (working name) is a household food planner that cuts grocery and cooked-food waste by telling a family *how much to buy* and *what to cook first*, based on how that family actually eats.

Differentiators: prevents waste at the shop (not after), plans for the whole family, tracks cooked food as well as groceries, estimates expiry for items with no printed date, and needs no daily logging.

Business model: free for families; kirana shops pay for digital bills and predicted customer reorders.

2. Problem statement

Working Indian families waste food they paid for because they buy more than they can eat before it spoils, and nobody tracks loose vegetables, milk, grains or leftovers that have no printed expiry date.

Evidence: UNEP's Food Waste Index Report 2024 puts India's household food waste at 55 kg per person per year, and links hotter climates to higher household waste.

Ground insights from an experienced homemaker:

- Grocery quantities must follow the family: number of members, meals per day, portion size, diet and each person's needs (e.g. ajwain for digestion).
- These decisions are made at **buying time**, so that is where the app must help.
- Food lasts differently by type: some grains keep a year, others 6 months, 1 month, 15 days, 8 days or 1 day. Dry, wet, liquid and long-life items need different planning.
- Cooked food lasts differently by preparation: roasted longest, raw sprouts 1–2 days, then fried, and cooked wet food is best eaten within hours.
- Earlier a full-time homemaker did this planning; today working women and men cannot plan perfectly at home and at work. Good planning lets them succeed at both.

3. Objectives and success criteria

The objective is a working prototype that cuts household food waste without causing shortages, proven in a 50-household, 90-day simulation calibrated to the UNEP 55 kg benchmark, and built to meet every judging criterion below.

Goal	Metric	Hackathon target
Cut food waste	kg wasted per household per month, baseline vs AnnaSaathi	Clear reduction, shown at 100%, 70% and 50% advice-following
Save money	₹ wasted per household per month	Shown alongside kg
Do not cause shortages	Days an item ran out before the next shop	No increase vs baseline
Low effort	Time to add a week's groceries	Under 30 seconds by bill scan
Accurate input	Bill lines read correctly	8 of 10 sample bills correct

How the prototype meets the judging criteria

Criterion	How AnnaSaathi meets it	Proof shown to judges
Working prototype and implementation quality	The full loop runs end to end on real bills; only P0 features until the loop is stable; each step tested	The 2-minute live demo, run without switching to slides
Innovation	Buy-right quantities at purchase time; cooked-food freshness tracking; expiry for undated food (tier × storage × weather); kirana QR bill	The same bill shown as a plain expiry list vs AnnaSaathi's buy-right list
Technical excellence	Schema-checked LLM extraction; explainable formulas with a reason on every suggestion; per-family learning; reproducible simulator	Reason line on each item; bill-scan accuracy on 10 real bills

Criterion	How AnnaSaathi meets it	Proof shown to judges
Practical impact	Lower kg and ₹ wasted with no rise in shortages; Marathi/Hindi voice; no daily logging	Impact dashboard with the 100% / 70% / 50% advice-following slider
Scalability	Item catalogue, shelf-life tiers and languages stored as data, not code; stateless API; kirana-led growth	Adding a new item or language by editing data only
Live demonstration	Preloaded demo household, demo mode that works without slow APIs, recorded backup video	Rehearsed script in section 9

4. Target users

User	Situation	Main need
Working parent (primary), e.g. Sneha, 38, Pune	Full-time job; family of 5; weekly kirana shop plus app top-ups	Exact buying quantities and a daily "cook this first" nudge, with no logging
Home cook / elder	Cooks daily; prefers Marathi; not comfortable typing	Voice input and a simple, large-text view
Kirana owner (business customer)	Bills on paper or a basic app; competes with quick-commerce	Loyal customers and predictable reorders

5. Scope

In scope (hackathon MVP): one household, the full weekly loop (profile → bill scan → inventory → cook-first plan → buy-right list), and the impact simulator.

Out of scope: login and accounts, calorie or medical diet advice, barcode database, general chatbot, social features, restaurants and canteens, offline mode.

6. Solution and key features

AnnaSaathi runs one weekly loop; the buy-right list prevents waste, and a 10-second weekly check keeps quantities accurate without daily logging.



AnnaSaathi weekly loop · 6 steps

Feature	Priority	Acceptance criteria
Family profile (members, meals, diet, preparation style)	P0	Set up in under 2 minutes; drives all quantities
Bill / order-screenshot scan	P0	8 of 10 sample bills read correctly; any line fixable in one tap
Inventory with shelf-life tiers and "days left" range (adjusted for storage and weather)	P0	Every item has a tier and a range; sorted by urgency
Cook-first daily plan	P0	2–3 Indian dishes using the most at-risk items, matching diet and preparation style
Buy-right shopping list	P0	Each item shows quantity and a one-line reason; nothing already in stock
Impact dashboard (simulator results)	P0	Baseline vs AnnaSaathi in kg and ₹, with an advice-following slider
Weekly 10-second check	P1	Swipe: still have / finished / thrown; updates use rates

Feature	Priority	Acceptance criteria
Voice input in Marathi / Hindi / English	P1	A spoken purchase adds the right items
Cooked-food "finish by" reminder	P1	Logging a dish shows a finish-by time
Kirana QR digital bill	P2 (mock)	Scanning the QR imports items with no OCR

Buy-right rule: quantity to buy = daily use × the shorter of shelf life and days until next shop − stock at home, rounded to pack size.

7. Technical approach and scalability

The LLM reads messy input and speaks the user's language; buying and cooking decisions come from explainable models.

- **Bill extraction:** vision LLM with a fixed JSON format → items, quantity, price; matched to a catalogue of ~150 Indian items with Marathi/Hindi names.
- **Shelf life:** tier lookup × storage factor × weather factor (free weather API), corrected by each family's feedback.
- **Consumption:** per-person portion × members × diet, updated from purchase gaps and the weekly check.
- **Cook-first:** items ranked by urgency; recipes from a public Indian recipe dataset scored by how many at-risk items they use.
- **Simulator:** 50 synthetic households × 90 days, baseline vs AnnaSaathi.
- **Stack:** Next.js mobile-first web app + FastAPI (Python) + SQLite; demo data preloaded.

Built to scale beyond the demo

- **Data, not code:** items, shelf-life tiers, recipes and language names live in tables, so a new region, item or language needs no code change.
- **Stateless API:** the backend keeps no session state, so more servers can be added as users grow; SQLite moves to PostgreSQL with no change to app logic.
- **Per-household learning:** each family's use rates and corrections are stored separately, so accuracy improves per home as data grows.
- **Low-cost AI use:** the LLM is called once per bill or voice note; all daily planning runs on cheap formulas, so cost stays low as users grow.

- **Growth channel:** one kirana shop brings its regular customers; the QR bill scales with the shop network, not with ads.

8. Assumptions, constraints and risks

Assumptions: most purchases come with a printed bill or digital order; simulator portion sizes are realistic estimates; families follow the advice only part of the time.

Constraints: 24 hours, one experienced developer and three beginners; impact is simulated, not measured in real homes; shelf-life values must be checked against reliable sources.

Risk	Mitigation
Families stop updating and the inventory goes wrong	Consumption is estimated; only a 10-second weekly check
A freshness estimate is wrong	Every estimate is a range with "check look and smell; when in doubt, throw it out"; printed dates override; cooked-food limits use the stricter of habit and official guidance (FSSAI, USDA)
Bill scan fails in the live demo	Preloaded scanned bills and a recorded backup video
Judges see "just another expiry app"	Lead the pitch with buy-right, cooked food and the simulator
Health or personal data misuse	Health needs stored only as family-entered preferences, no medical advice; personal details removed from demo bills

9. Development approach and live demo

One full loop must work by hour 12; features freeze at hour 18 and code freezes at hour 21.

Person	Role	Owns
Harsh Kale	Engineering lead	App, bill extraction, models, buy-right and cook-first logic, integration

Person	Role	Owns
Soham Fulsundar	Data and simulator	Item catalogue with Marathi/Hindi names, shelf-life table with sources, household simulator, testing
Gajanan Bhakde	Evidence and kirana	Real bills (personal details removed), mock kirana QR, assumptions page, README, timekeeping
Vaishnavi Shirole	Story and user experience	Pitch, slides, screen wording, demo script, backup video

Hours	Milestone
0–1	Scope locked, repo and CLAUDE.md ready
1–6	Bill scan → inventory screen; simulator v1
6–12	Shelf life, cook-first and buy-right working; **full loop end to end**
12–18	Impact dashboard, polish, P1 features only if stable; **feature freeze**
18–21	Bug fixes, backup video, slides final; **code freeze**
21–24	Pitch rehearsal, at least 3 times

Quality rules while building

- Build the loop thin first, then deepen it; never start a P1 feature while a P0 step is broken.
- Test bill scanning on a fixed set of 10 real bills and track accuracy after every change.
- Unit-test the buy-right and urgency formulas with hand-checked examples.
- Run the simulator with a fixed random seed so results are the same every time.
- Commit small and often from all four members.

Live demo script (2 minutes)

1. Open the Patil family profile: 5 members, two working parents, an elder who prefers roasted food.
2. Scan a real kirana bill; items appear with tiers and days left.

3. Show today's cook-first plan using the most at-risk items.
4. Show the buy-right list: "Tomatoes: buy 1 kg, not 2," with its reason.
5. Show the impact dashboard and move the advice-following slider.

Fallbacks: preloaded scanned bills if the camera or API fails; a recorded video of the same script.

10. Sources

- UNEP Food Waste Index Report 2024 figures, as reported by Drishti IAS and PW Only IAS
- Triwizathon 2026, Problem Statement 5
- Homemaker insights shared by a team member's parent (24 Oct 2026)